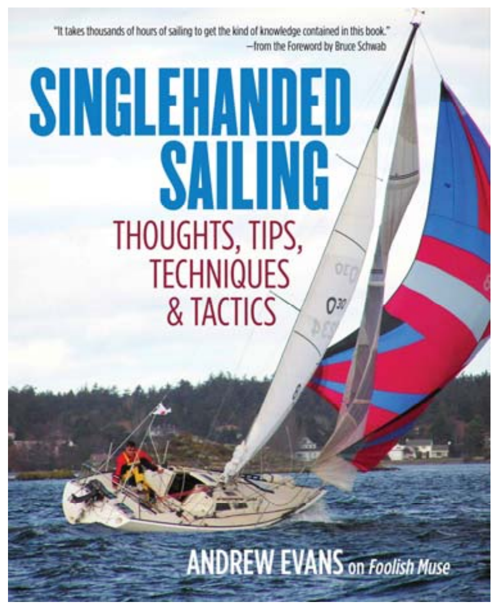
# Better Broaching for Singlehanders

A supplement to Singlehanded Sailing: Thoughts, Tips, Techniques & Tactics

Being a very aggressive singlehanded sailor, I'm pretty sure that I have suffered MANY more spinnaker broaches than most people would like to admit.  I thought it would be good to jot down a few of the things I've learned.

First and foremost, don't be afraid of a broach.  Unless you don't fly your spinnaker at all, you can't expect to eliminate broaches.  They are what we do.  Just enjoy them as part of the adventure.

Second, when you watch video of crewed boats broaching, it often looks like a bunch of headless chickens running around.  As a singlehander, you can remain calm, cool and collected through the whole process and come out smiling at the end.  So don't worry, be happy.



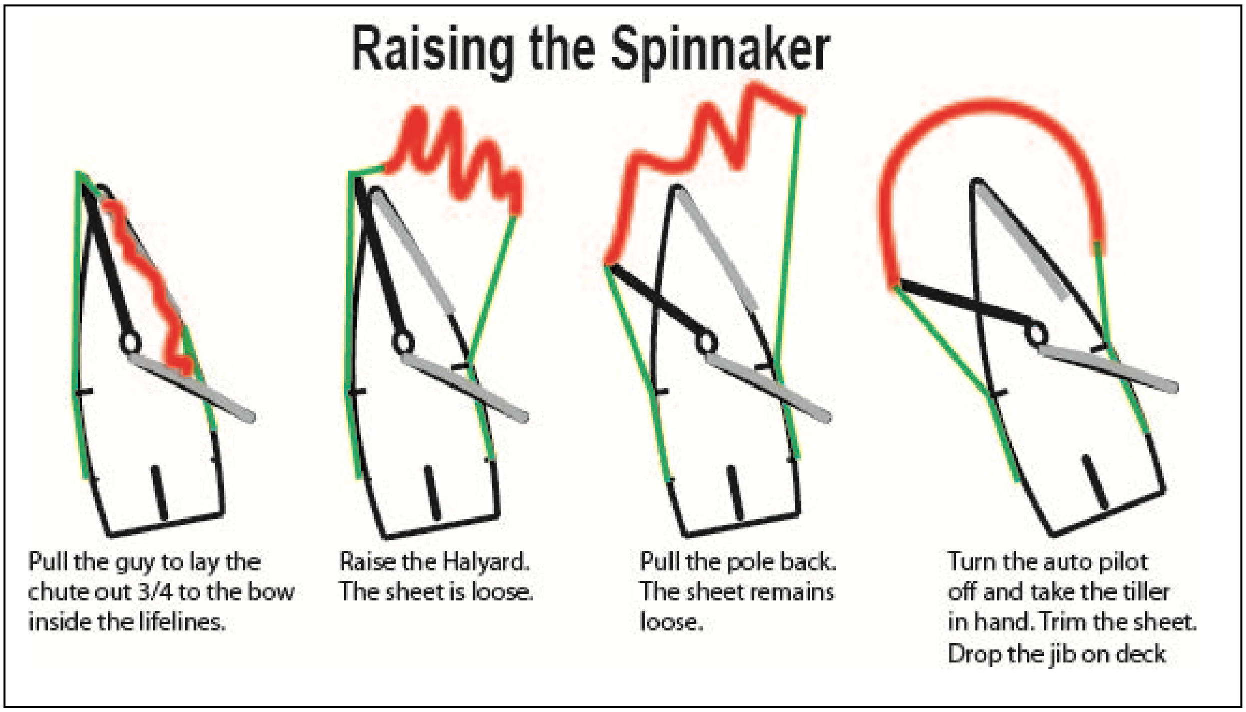

**Singlehanded Sailing**

With over 12,000 copies sold around the world, this is a great reference for singlehanded performance sailing.

**Boat design:** Your boat will be a major factor in how often you broach.  Heavy cruisers or full keeled boats will broach less often than short keel, small rudder boats like my Olson 30.  But that really just means that cruisers should be flying the chute in higher winds.  And, on the other hand, full keel boats will have more trouble turning back down wind once they do broach. It's all a tradeoff.

**Avoiding a broach in the first place:** I had a chuckle watching a famous skipper's instructional video on the topic.  When a broach was imminent, she calmly asked her crew to ease the spinnaker sheet and boom vang.  The crew calmly complied and the broach was avoided.  In my experience, a broach happens so fast that it would be near impossible to even think and speak a command, let alone have the crew comply.  This is where singlehanders have an advantage, with our hand on the tiller we can feel the broach start and immediately take steps to avoid it, without all the pesky giving of orders.  When I'm out for an afternoon sail, I'll be sitting on the high side of the boat with the spinnaker sheet crossing the cockpit and on the winch next to me.  So if the boat starts to broach, I can ease the sheet very quickly, certainly much quicker than a crew could do it.

**Launching the Spinnaker.** I've found that my autopilot can't manage the sudden jolt and change in speed of the spinnaker filling in higher winds. I had many broaches before I figured out how to avoid them on launch. What I do is make sure the spinnaker sheet is eased by about 12' before I pull up the halyard, so when the spinnaker is first raised, it is flogging in the wind and does not fill. Then I pull back the pole, turn off the autopilot and take the tiller in hand. Finally I pull in the sheet until the spinnaker fills. On my boat this is necessary in anything above 15 knot winds. At 25 knots it's absolutely critical.




**Spinnaker setup:** The cause of a broach while sailing is that your spinnaker has much more power than your rudder. Unless the spinnaker is properly trimmed for the wind angle, the rudder will not be able to keep the boat underneath the spinnaker. So the first step to avoiding a broach is to set up the spinnaker for the apparent wind direction. Don't ever think of the spinnaker as just a sack that the wind pushes. The spinnaker, like all of your sails, is a foil with a proper entry and exit. And just as you change the position of your jib or mainsail on a reach, the setup of the spinnaker is vital on a run. (Think about this for a moment. If you are sailing on a beam reach with the wind at 90°, and if you pull your main and jib in tight, then the boat will round up into the wind. The same holds for a spinnaker.)

You want the pole to be perpendicular (at 90°) to the wind, or somewhere near to that. You can probably move the pole forward to 110°, but no further. I do not accept the idea of hiding the spinnaker behind the mainsail. The reason is that the shape of the spinnaker will want to pull the boat up to its correct position relative to the wind, and this will lead to a broach by itself. In higher winds, I'll generally sail no deeper than 150-160 and I'll have the spinnaker pole pulled back from 1/2 to 2/3. This gives me good control.

In this photo I was just about to broach on a beam reach, so I eased the sheet until the sail was flogging. This stopped the broach. (Note that I had previously eased the halyard to undo an hourglass – but that's a different issue.)

It might happen that with the wind so high, I can't grind the pole back as far as I'd like. (Especially if I don't have an autopilot to take over steering.) If this is the case, then I'll steer by hand and keep as much control as possible. But this is one of the reasons why I've broached so often. The important key is to keep the spinnaker pole, and the luff of the spinnaker as close as possible to perpendicular to the wind.

**Ease the sheet:**

The next step is to ensure that the spinnaker sheet is eased until the luff is just at the point of curl. Only if you are manually steering and have the sheet in hand should you use the crewed boat technique of sheeting and easing each time the spinnaker starts to curl. Otherwise you are just leading to a broach.

If you are using an autopilot or wind vane, you want the chute to collapse or partially collapse any time that the boat starts to head up. Even if you can pull the sheet by hand, <u>do not do this</u> if the autopilot is steering. It merely tells the boat to swing windward, into the broach you're trying to avoid. Rather, allow the autopilot to steer the boat back downwind and the chute will refill by itself. Only pull in the sheet if you have turned off the autopilot and have the tiller in your hand.

Very often when I'm sailing, I'll watch as the first few feet of the luff curls over on itself and the autopilot steers the boat back down until the luff uncurls. I know that this is the best sail trim. If you have the sheet so tight that the luff never curls over, then you are leading to a broach.


A boat is quite stable on its side when broaching

Sometimes the spinnaker will collapse completely. This is just a sign that the boat needs to round down a lot. So don't sheet in. Let the autopilot take the boat down, or disconnect the autopilot and hand steer down.

**Once the Broach happens:**

Once you start flying a chute in higher winds, then broaching is inevitable. As I said in the first section, don't worry - relax. A famous IMOCA 60 skipper told me that when things go bad, "Make a cup of tea, and drink it. And if you're French then smoke a cigarette." I won't say that I've followed his advice to make a cup of tea, but I have taken sips from the cup next to me in the cockpit while my boat was on its side. That's when I knew I was getting the hang of it. (I'll admit that it took me about 50 broaches before I reached this level of calm.)

This is the first thing that will happen if you release the guy when broached. Don't do it! There is no way to recover.

**The first rule of broaching** is do **NOT** release the guy or the halyard. Doing either of these will cause a lot of problems that end of with your spinnaker in the water, dragging beside or behind the boat. I've done these things numerous times in desperation, and I can confidently say that <u>every single time</u>, I've ended up with a very long rip in the spinnaker. I do my own sail repairs, so all it costs is a slightly smaller spinnaker.


The next bad step is to release the halyard. There is zero chance of recovery from this without putting their chute in the water.

Once you release the guy or the halyard, or both, you will never be able to recover the spinnaker. Even grinding in on the sheet will be impossible. I know this because I've tried it. There is far too much pressure on the sail.

The big problem is that once you have released the guy or the halyard, or both, then you don't have enough spinnaker sheet left for you to ease the sheet until the spinnaker is flogging. You are stuck with a full spinnaker under immense pressure. The only thing that you can do is blow the guy completely. This means that you have your spinnaker flying 60' out the side of your boat at the end of the sheet, then 20'-30' of spinnaker foot, and finally with the guy now flying another 60' further out. It's all stretching 120' out the side of your boat. Everything is flapping around and you can't pull the spinnaker back on board. You can't even grind it in with your winch. I have done this and after you pull the spinnaker in from the water, you end up with your guy twisted into a one cubic foot tangle of knots. It took me a half hour to untwist it.

My own boat (with a crew) doing exactly the wrong thing.

**The steps to take:**
What you want to end up with is your boat back upright, aiming in the right direction downwind, first with the spinnaker flogging in front, then with the spinnaker properly trimmed and sailing at full speed again.



The steps you take will depend on the wind speed.

In winds up to 15 knots, the only solution is to ease the spinnaker sheet until the spinnaker is flogging in front of the boat. The boat will come back upright. Then simply steer the boat downwind (either hand steer or let your autopilot do it) until you are back in control. Then just sheet in again and start sailing.

In winds 15-20 knots, easing the spinnaker sheet alone might not be enough to get the boat back upright and sailing downwind. The solution is to ease the mainsheet **first**, then ease the spinnaker sheet as above. Get the boat under control and sheet in again. As the winds get up, your autopilot might not have the intelligence to complete this. I've found it necessary to hand steer in these conditions. Make sure to ease the mainsheet first.

In winds above 20 knots, **first** pop the boom vang, **then** ease the main sheet, **then** the spinnaker sheet as above. To do this you must have the boom vang control in a convenient place. If the cleat is at the base of the mast, it will be very difficult for you to get up to release it. My boom vang line is on the boom, about four feet back from the mast. So it is easier to reach. Make sure to pop the boom vang first, then the main sheet, then the spinnaker sheet. It would be very difficult to pop the vang if your boom is dragging in the water on the low side of the boat.

**Don't be afraid to flog the spinnaker.** Your sails can handle the abuse and it really is the only way to get back upright and sailing again. So ease the spinnaker sheet a long way, until the sail is just floating in the wind ahead of the boat. Once you are up straight and pointing downwind, then pull in the sheet and start accelerating.

**Get sailing again.** You've been through a broach and recovered. There is no reason for you to not continue sailing with your spinnaker flying and you smiling. Your job is to get the boat sailing again at top speed. So after you've broached and recovered, don't quit. In most situations there is no reason to douse your spinnaker, so don't even consider it. After a bit of experience you will get to be expert at this process.

**If you decide to douse the spinnaker completely**
If you have decided that there is just too much wind to fly your spinnaker safely, then you can douse it completely. But you MUST go through very specific steps to get it down without a disaster. As I said earlier, if you are on your side in a broach, do not pop the guy or the halyard even if you have decided to douse the spinnaker completely. You must get the boat back upright again before you do any of these. So go through the process of easing your boom vang and mainsheet. However once you are upright and pointing in the right direction, you don't need to sheet in the spinnaker. You can start the douse procedure as written below.

**How to douse the spinnaker in high winds.**
I've been sailing with no autopilot or wind vane for several years, so I've had to perfect the steps of getting a chute down in high winds. I have found that I can't keep the tiller between my knees while reaching over the leeward side to grab the spinnaker sheet if it is still tight and the boat is surfing at high speed. So what I've learned is to ease the sheet until the spinnaker is flogging in front and the boat has slowed to a manageable speed. Then I aim deep down wind. There is a big difference in control between surfing at 13 knots with the spinnaker up, and merely sailing at 7 knots when the spinnaker is flogging in front of the boat. It only takes a few seconds to for the speed to drop. Then I can grab the sheet in one hand and pop the guy with the other hand, then gather up as much of the foot as possible, then pop the halyard and pull like mad to get as much of the spinnaker into the cockpit as I can, minimizing the amount of sail in the water.

The singlehanders view of a chinese gybe.
In this case you do release the guy.

**A chinese gybe** is essentially the opposite of a broach. Instead of rounding up into the wind, the boat rounds down. So your spinnaker pole is in the water and then your mainsail flops over (hitting you in the head) and drags in the water as well. There is a pretty good chance of breaking the pole, or the uphaul, or the mast connection point.

The best way to avoid this is to make sure you don't sail too deep downwind. And especially don't sail by the lee. In high winds you don't want to sail deeper than 165° where the slightest wind shift will cause a gybe.

A chinese gybe can also happen if you have followed the advice from another paper of mine, and have intentionally gybed without shifting the pole for an extended period. The wind picks up and you are not paying close attention, and essentially you broach and the pole is in the water.

The solution to the chinese gybe is the same, just opposite as the steps for a broach. Remember that in this case, the key line is now the guy at the bottom of the boat running through your spinnaker pole.  So do not release the sheet on the top of the boat. Rather you must release the guy and let the sail flog in front of the boat, until you can get back upright. Then you gybe and end up with the pole on the windward side, and pull the guy in again and immediately ease the sheet. Of course the spinnaker is going to get tangled up in your moves to get upright. But all the rules remain the same. Don't pop the halyard and don't pop the windward sheet until back upright. And even then, don't try to douse the sail until you are back with the pole on the windward side. It all ends up a mess if you try anything else.

Here is a great video of a chinese gybe on a singlehanded 60' IMOCA singlehanded boat: https://www.youtube.com/watch?v=f14r3tC-Y4E

**What to do when you've done everything wrong?**

I wish I had advice for what to do after you have blown the guy or the halyard. Your spinnaker will certainly end up in the water dragging behind your boat. Of course your boat is still moving at speed in whatever direction it happens to be pointing so you will have to work extremely hard to drag the soaking wet spinnaker back on board. I've even had to turn the boat around because I was aiming at some rocks. I have tried releasing the guy completely so it is dragging far behind, but this does not make it any easier to pull the chute on board, and it does not reduce the chance of ripping your spinnaker for 20' or more.  I have not tried cutting the halyard free and letting it run through the mast. Perhaps someone else will try this.  But just remember my second paragraph in this paper. We are athletes and we are out sailing when other poor slobs are stuck at home cutting the lawn. Be happy and enjoy the adventure. Learn to use a sewing machine and it will all come out all right in the end.

All that's left of the spinnaker after a particularly bad broach on my boat caused by a super moon. Even a sewing machine couldn't save this

This is what your spinnaker and guy will look like if you blow the guy in a broach. This broach was caused by hitting a huge kelp bed at high speed.